CHAPTER 2

Physical Characteristics of Crude Oil

Understanding the fundamental physical properties that define every barrel — and decide how it is refined, priced and traded.

NO TWO BARRELS ARE ALIKE

8°–45°+

the API-gravity range of the world's crudes

"Crude oil" is not one substance but hundreds of distinct grades. What separates a $5 premium from a $20 discount is rarely where it comes from — it is the physical chemistry of the barrel itself.

Two master properties

Density (API gravity) and sulfur content do most of the work in classifying and pricing a crude.

Properties set the refining path

They decide which products a barrel yields and how much processing it takes to get there.

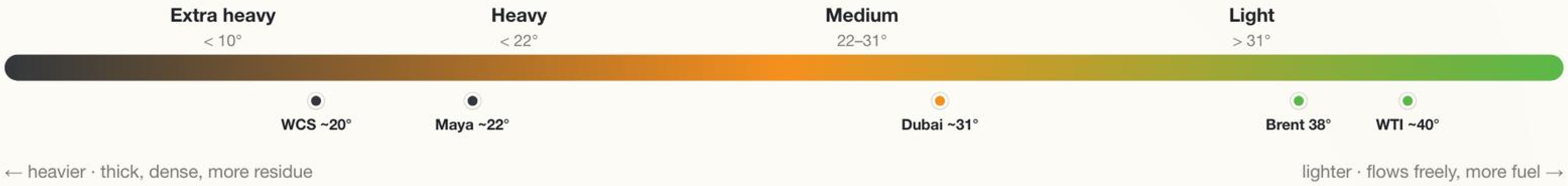
Properties set the price

Every grade trades at a premium or discount to a benchmark, driven first by its physical quality.

PROPERTY 1 · DENSITY

API gravity — light vs heavy

API gravity measures how light a crude is relative to water. The higher the number, the lighter the oil — and the more valuable light products it naturally yields.



Light crude is premium

High API barrels yield more gasoline, diesel and jet with less processing — so they command the highest prices.

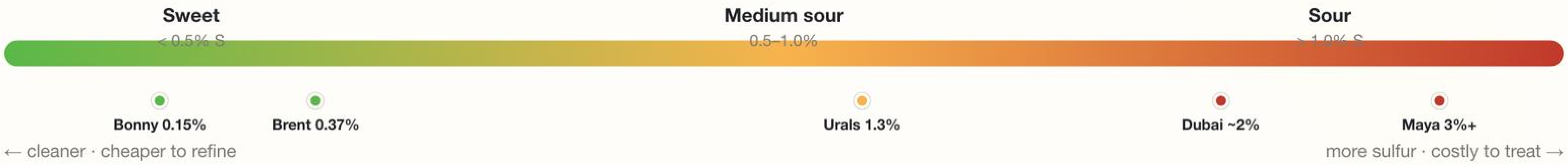
Heavy crude is discounted

Low API barrels are dense and rich in residue, needing complex, costly refining — so they trade cheaper.

PROPERTY 2 · SULFUR

Sulfur content — sweet vs sour

Sulfur is the main impurity in crude. Low-sulfur "sweet" crude is cleaner and easier to process; high-sulfur "sour" crude needs extra treating to meet fuel specifications.



Sweet = simpler & cleaner

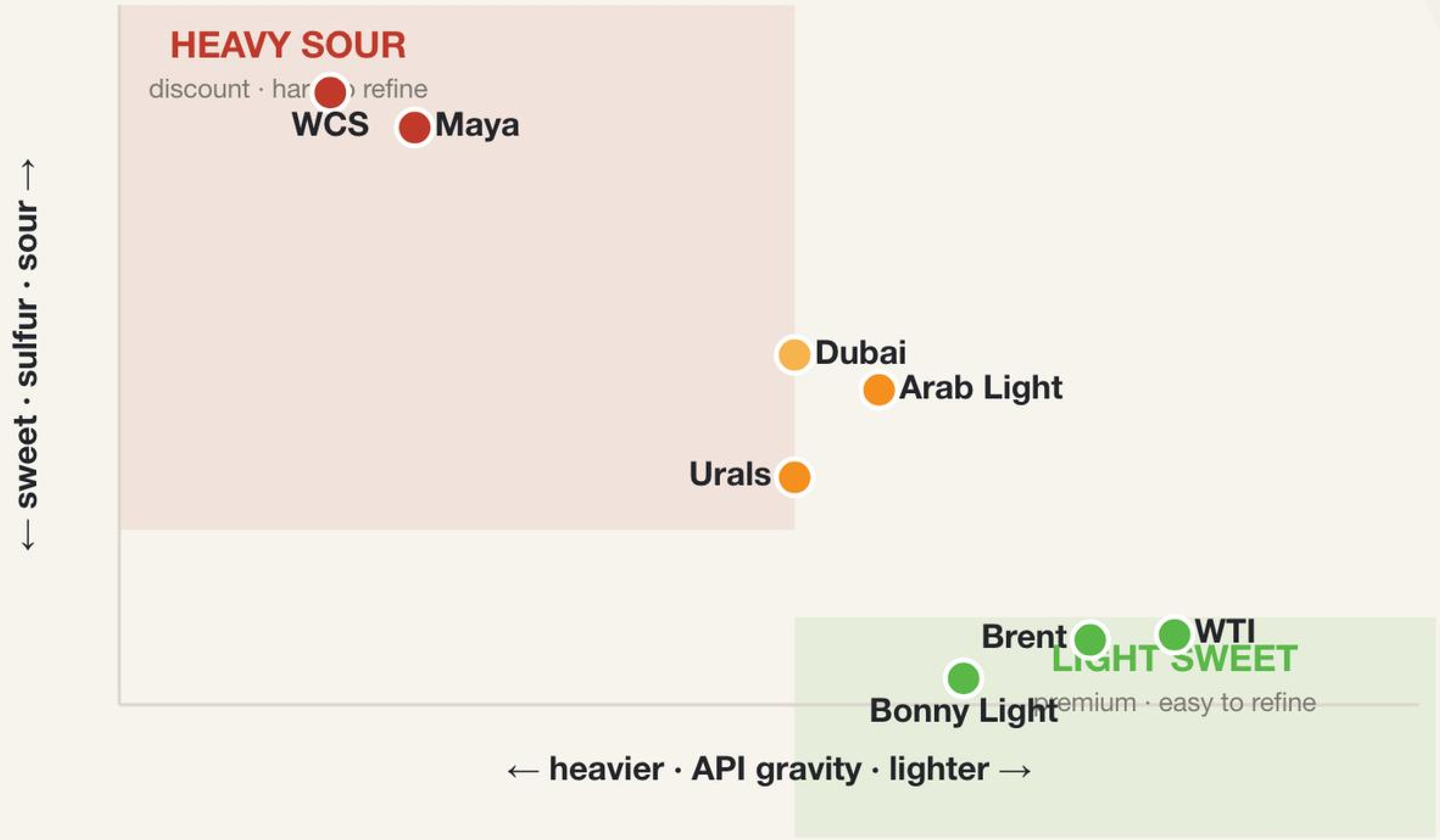
Low sulfur means easier compliance with clean-fuel rules and less investment in treating units.

Sour = harder & dirtier

High sulfur must be removed to make road fuels, so sour crude needs complex refineries and sells at a discount.

THE BIG PICTURE

The crude classification map



Two axes, one quality

Plot any crude by density and sulfur and its value almost falls out of the chart.

Opposite corners

The benchmarks — WTI, Brent — sit in the prized light-sweet corner; Maya and WCS in the discounted heavy-sour one.

BEHAVIOUR

Viscosity and cold flow

Viscosity

How easily crude flows. Light crude is thin and pumps readily; heavy crude is thick and may need heating or dilution to move through a pipeline.

Pour point

The lowest temperature at which crude still flows. Waxy crudes can solidify in cold pipelines if it drops too far.

Cloud point

The temperature at which wax crystals first appear, clouding the oil — an early warning before flow is lost.

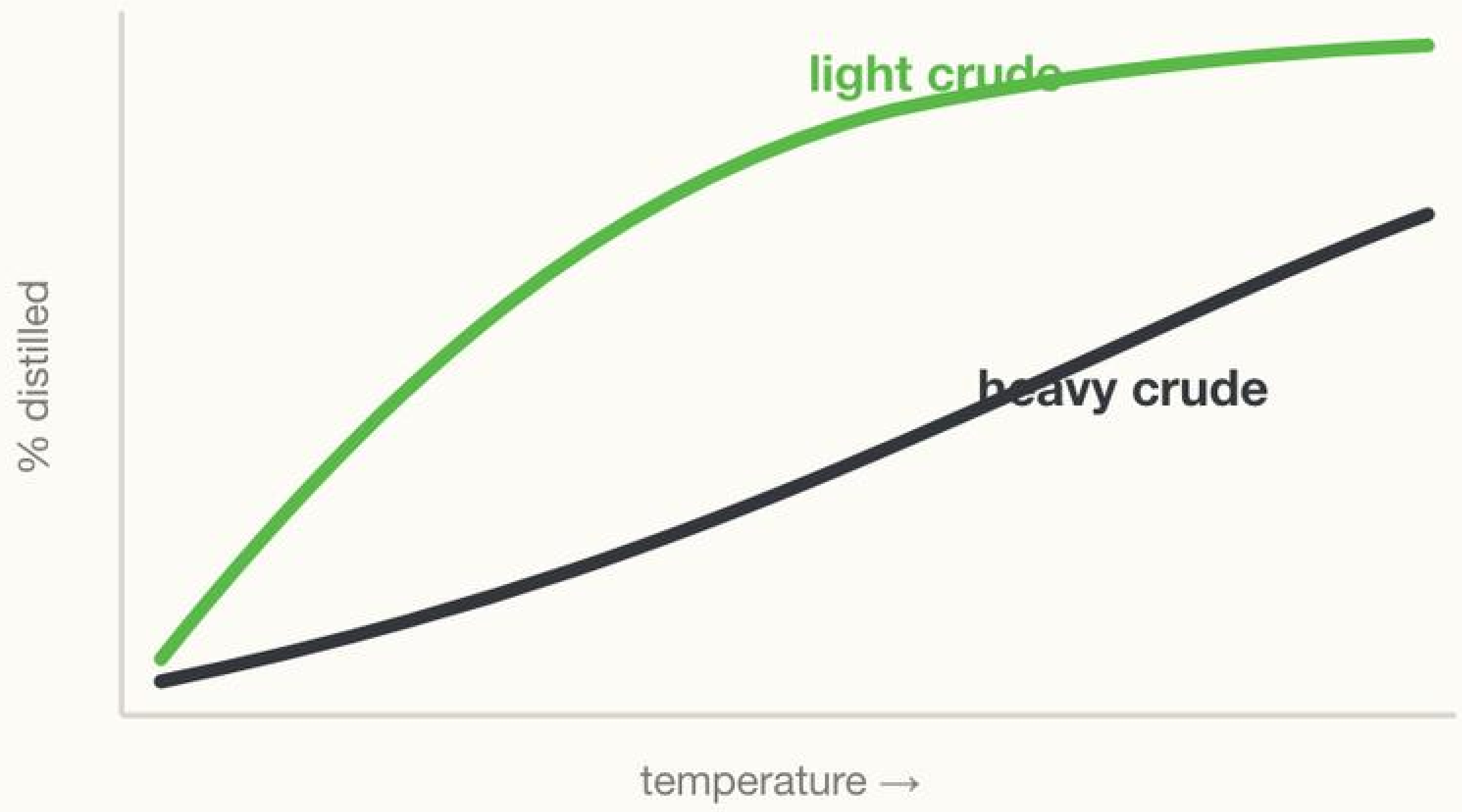
Why it matters

Flow properties decide how crude is stored, pumped and shipped — and what it costs to handle far from where it was produced.

WHAT IT YIELDS

Distillation behaviour

DISTILLATION CURVE — % BOILED OFF VS TEMPERATURE



Light crude boils early

Most of it vaporises at low temperatures, giving a large cut of valuable light products — petrol, naphtha, jet.

Heavy crude lags

Much stays liquid until high temperatures, leaving heavy gasoil and residue that must be cracked to add value.

The yield is set at the well

A barrel's natural product slate is largely fixed by its chemistry — refining only reshapes it.

THE FINER DETAIL

Other key properties

Acidity (TAN)

Total Acid Number measures naphthenic acids that corrode refinery equipment — high-TAN crudes trade at a discount.

Metals & contaminants

Vanadium and nickel poison refinery catalysts; their level limits how a crude can be processed.

Salt & water (BS&W)

Basic sediment and water must be low — buyers reject cargoes that carry too much non-oil ballast.

Volatility (RVP)

Reid Vapour Pressure gauges how readily a crude gives off vapour — key for safe storage and transport.

FROM CHEMISTRY TO PRICE

Why properties drive value

PREMIUM

Light & sweet

More valuable products, less processing, easy compliance. Refined in almost any plant — so it sells at a premium to benchmark.

QUALITY

→

sets the differential

DISCOUNT

Heavy & sour

Cheaper feedstock, but only complex refineries can process it profitably — so it sells at a discount to benchmark.

Every grade is priced as a differential to a benchmark like Brent or WTI — and that differential is set first by the physical properties in this chapter.

CONCLUSION — THE DNA OF A BARREL

Physical properties are the DNA of every barrel.

Density & sulfur lead

API gravity and sulfur do most of the work — light-sweet to heavy-sour is the master scale.

Chemistry sets the yield

Distillation, viscosity and contaminants decide what a barrel becomes and how it is handled.

Properties become price

Every grade's premium or discount to benchmark traces back to the physics in this chapter.